# ICPC Notebook

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# template

## .python-version

```
pypy-3.11.13
```

## hash.py

md5: 14abff

```python
# 使い方: uv run hash.py -> コピペ -> Ctrl + D
# コメント・空白・改行を無視して AST ベースで md5 ハッシュする
import ast, hashlib, sys

tree = ast.parse(sys.stdin.read())
dump = ast.dump(tree)
print(hashlib.md5(dump.encode()).hexdigest()[:6], end="")
```

## settings.sh

```
# CLion の設定
Settings → Build → CMake → Reload CMake Project
add_compile_options(-D_GLIBCXX_DEBUG)
# Caps Lock を Ctrl に変更
setxkbmap -option ctrl:nocaps
```

## template.py

md5: 61e08b

```python
import sys
input = sys.stdin.readline
sys.setrecursionlimit(1 << 20)

# your code here...
```

# data-structure

## BIT.hpp

```cpp
struct BIT {
   vector<ll> a;
   BIT(ll n) : a(n + 1) {}
   void add(ll i, ll x) {  // A[i] += x
      i++;
      while(i < sz(a)) {
         a[i] += x;
         i += i & -i;
      }
   }
   ll sum(ll r) {
      ll s = 0;
      while(r) {
         s += a[r];
         r -= r & -r;
      }
      return s;
   }
   ll sum(ll l, ll r) {  // sum of A[l, r)
      return sum(r) - sum(l);
   }
};
```

## FastSet.hpp

```cpp
// using u64 = uint64_t;
const u64 B = 64;
struct FastSet {
   u64 n;
   vector<vector<u64>> a;
   FastSet(u64 n_) : n(n_) {
      do a.emplace_back(n_ = (n_ + B - 1) / B);
      while(n_ > 1);
   }
   // bool operator[](ll i) const { return a[0][i / B] >> (i %
B) & 1; }
   void set(ll i) {
      for(auto& v : a) {
         v[i / B] |= 1ULL << (i % B);
         i /= B;
      }
   }
   void reset(ll i) {
      for(auto& v : a) {
         v[i / B] &= ~(1ULL << (i % B));
         if(v[i / B]) break;
         i /= B;
```

```
      }
   }
   ll next(ll i) {  // i を超える最小の要素
      rep(h, 0, sz(a)) {
         i++;
         if(i / B >= sz(a[h])) break;
         u64 d = a[h][i / B] >> (i % B);
         if(d) {
            i += countr_zero(d);
            while(h--) i = i * B + countr_zero(a[h][i]);
            return i;
         }
         i /= B;
      }
      return n;
   }
   ll prev(ll i) {  // i より小さい最大の要素
      rep(h, 0, sz(a)) {
         i--;
         if(i < 0) break;
         u64 d = a[h][i / B] << (~i % B);
         if(d) {
            i -= countl_zero(d);
            while(h--) i = i * B + __lg(a[h][i]);
            return i;
         }
         i /= B;
      }
      return -1;
   }
};
```

# math

## BinaryGCD.hpp

```
u64 ctz(u64 x) { return countr_zero(x); }
u64 binary_gcd(u64 x, u64 y) {
   if(!x || !y) return x | y;
   u64 n = ctz(x), m = ctz(y);
   x >>= n, y >>= m;
   while(x != y) {
      if(x > y) x = (x - y) >> ctz(x - y);
      else y = (y - x) >> ctz(y - x);
   }
   return x << min(n, m);
}
```

## ExtGCD.hpp

```
// returns gcd(a, b) and assign x, y to integers
// s.t. ax + by = gcd(a, b) and |x| + |y| is minimized
ll extgcd(ll a, ll b, ll& x, ll& y) {
   // assert(a >= 0 && b >= 0);
   if(!b) return x = 1, y = 0, a;
   ll d = extgcd(b, a % b, y, x);
   y -= a / b * x;
   return d;
}
```

# modint

## BarrettReduction.hpp

```
// using u64 = uint64_t;
struct Barrett {  // mod < 2^32
   u64 m, im;
   Barrett(u64 mod) : m(mod), im(-1ULL / m + 1) {}
   // input: a * b < 2^64, output: a * b % mod
   u64 mul(u64 a, u64 b) const {
      a *= b;
      u64 x = ((__uint128_t)a * im) >> 64;
      a -= x * m;
      if((ll)a < 0) a += m;
      return a;
   }
};
```

## modint.hpp

```
const ll mod = 998244353;
struct mm {
   ll x;
   mm(ll x_ = 0) : x(x_ % mod) {
      if(x < 0) x += mod;
   }
   friend mm operator+(mm a, mm b) { return a.x + b.x; }
   friend mm operator-(mm a, mm b) { return a.x - b.x; }
   friend mm operator*(mm a, mm b) { return a.x * b.x; }
   friend mm operator/(mm a, mm b) { return a * b.inv(); }
   // 4 行コピペ  Alt + Shift + クリックで複数カーソル
   friend mm& operator+=(mm& a, mm b) { return a = a.x + b.x; }
   friend mm& operator-=(mm& a, mm b) { return a = a.x - b.x; }
   friend mm& operator*=(mm& a, mm b) { return a = a.x * b.x; }
   friend mm& operator/=(mm& a, mm b) { return a = a * b.inv();
}
   mm inv() const { return pow(mod - 2); }
   mm pow(ll b) const {
      mm a = *this, c = 1;
      while(b) {
         if(b & 1) c *= a;
         a *= a;
         b >>= 1;
      }
      return c;
   }
};
```

# FPS

## FFT.hpp

```
// {998244353, 3}, {1811939329, 13}, {2013265921, 31}
mm g = 3;  // 原始根
void fft(vector<mm>& a) {
   ll n = sz(a), lg = __lg(n);
   assert((1 << lg) == n);
   vector<mm> b(n);
   rep(l, 1, lg + 1) {
      ll w = n >> l;
      mm s = 1, r = g.pow(mod >> l);
      for(ll u = 0; u < n / 2; u += w) {
         rep(d, 0, w) {
            mm x = a[u << 1 | d], y = a[u << 1 | w | d] * s;
            b[u | d] = x + y;
            b[n >> 1 | u | d] = x - y;
         }
         s *= r;
      }
      swap(a, b);
   }
}
vector<mm> conv(vector<mm> a, vector<mm> b) {
   if(a.empty() || b.empty()) return {};
   size_t s = sz(a) + sz(b) - 1, n = bit_ceil(s);
   // if(min(sz(a), sz(b)) <= 60) 愚直に掛け算
   a.resize(n);
   b.resize(n);
   fft(a);
   fft(b);
   mm inv = mm(n).inv();
   rep(i, 0, n) a[i] *= b[i] * inv;
   reverse(1 + all(a));
   fft(a);
   a.resize(s);
   return a;
}
```

## FFT_fast.hpp

```
// modint を u32 にして加減算を真面目にやると速い
mm g = 3;  // 原始根
void fft(vector<mm>& a) {
   ll n = sz(a), lg = __lg(n);
   static auto z = [] {
      vector<mm> z(30);
      mm s = 1;
      rep(i, 2, 32) {
         z[i - 2] = s * g.pow(mod >> i);
         s *= g.inv().pow(mod >> i);
      }
      return z;
   }();
   rep(l, 0, lg) {
      ll w = 1 << (lg - l - 1);
      mm s = 1;
      rep(k, 0, 1 << l) {
         ll o = k << (lg - l);
         rep(i, o, o + w) {
```

```
            mm x = a[i], y = a[i + w] * s;
            a[i] = x + y;
            a[i + w] = x - y;
         }
         s *= z[countr_zero<uint64_t>(~k)];
      }
   }
}
// コピペ
void ifft(vector<mm>& a) {
   ll n = sz(a), lg = __lg(n);
   static auto z = [] {
      vector<mm> z(30);
      mm s = 1;
      rep(i, 2, 32) {  // g を逆数に
         z[i - 2] = s * g.inv().pow(mod >> i);
         s *= g.pow(mod >> i);
      }
      return z;
   }();
   for(ll l = lg; l--;) {  // 逆順に
      ll w = 1 << (lg - l - 1);
      mm s = 1;
      rep(k, 0, 1 << l) {
         ll o = k << (lg - l);
         rep(i, o, o + w) {
            mm x = a[i], y = a[i + w];  // *s を下に移動
            a[i] = x + y;
            a[i + w] = (x - y) * s;
         }
         s *= z[countr_zero<uint64_t>(~k)];
      }
   }
}
vector<mm> conv(vector<mm> a, vector<mm> b) {
   if(a.empty() || b.empty()) return {};
   size_t s = sz(a) + sz(b) - 1, n = bit_ceil(s);
   // if(min(sz(a), sz(b)) <= 60) 愚直に掛け算
   a.resize(n);
   b.resize(n);
   fft(a);
   fft(b);
   mm inv = mm(n).inv();
   rep(i, 0, n) a[i] *= b[i] * inv;
   ifft(a);
   a.resize(s);
   return a;
}
```

# graph

# graph/tree

# flow

## 燃やす埋める.md

| 変形前の制約 | 変形後の制約 |
|---|---|
| $x$ が $0$ のとき $z$ 失う | $(x, T, z)$ |
| $x$ が $0$ のとき $z$ 得る | 無条件で $z$ 得る; $(S, x, z)$ |
| $x$ が $1$ のとき $z$ 失う | $(S, x, z)$ |
| $x$ が $1$ のとき $z$ 得る | 無条件で $z$ 得る; $(x, T, z)$ |
| $x, y, \ldots$ がすべて $0$ のとき $z$ 得る | 無条件で $z$ 得る; $(S, w, z), (w, x, \infty), (w, y, \infty)$ |
| $x, y, \ldots$ がすべて $1$ のとき $z$ 得る | 無条件で $z$ 得る; $(w, T, z), (x, w, \infty), (y, w, \infty)$ |

# string

## KMP.hpp

```
// kmp[i] := max{ l ≤ i | s[:l] == s[(i+1)-l:i+1] }
// abacaba -> 0010123
auto KMP(string s) {
   vector<ll> p(sz(s));
   rep(i, 1, sz(s)) {
      ll g = p[i - 1];
      while(g && s[i] != s[g]) g = p[g - 1];
      p[i] = g + (s[i] == s[g]);
   }
   return p;
}
```

## Manacher.hpp

```
// 各位置での回文半径を求める
// aaabaaa -> 1214121
// 偶数長の回文を含めて直径を知るには, N+1 個の $ を挿入して 1 を引く
// $a$a$a$b$a$a$a$ -> 123432181234321
auto manacher(string s) {
   ll n = sz(s), i = 0, j = 0;
   vector<ll> r(n);
   while(i < n) {
      while(i >= j && i + j < n && s[i - j] == s[i + j]) j++;
      r[i] = j;
      ll k = 1;
      while(i >= k && i + k < n && k + r[i - k] < j) {
         r[i + k] = r[i - k];
         k++;
      }
      i += k, j -= k;
   }
   return r;
}
```

## RollingHash.hpp

```
// using u64 = uint64_t;
const u64 mod = INF;
u64 add(u64 a, u64 b) {
   a += b;
   if(a >= mod) a -= mod;
   return a;
}
u64 mul(u64 a, u64 b) {
   auto c = (__uint128_t)a * b;
   return add(c >> 61, c & mod);
}
random_device rnd;
const u64 r = ((u64)rnd() << 32 | rnd()) % mod;
struct RH {
   ll n;
   vector<u64> hs, pw;
   RH(string s) : n(sz(s)), hs(n + 1), pw(n + 1, 1) {
      rep(i, 0, n) {
         pw[i + 1] = mul(pw[i], r);
         hs[i + 1] = add(mul(hs[i], r), s[i]);
      }
   }
   u64 get(ll l, ll r) const { return add(hs[r], mod -
mul(hs[l], pw[r - l])); }
};
```

## SuffixArray.hpp

```
// returns pair{sa, lcp}
// sa 長さ n : s[sa[0]:] < s[sa[1]:] < … < s[sa[n-1]:]
// lcp 長さ n-1 : lcp[i] = LCP(s[sa[i]:], s[sa[i+1]:])
auto SA(string s) {
   ll n = sz(s) + 1, lim = 256;
   // assert(lim > ranges::max(s));
   vector<ll> sa(n), lcp(n), x(all(s) + 1), y(n), ws(max(n,
lim)), rk(n);
   iota(all(sa), 0);
   for(ll j = 0, p = 0; p < n; j = max(1LL, j * 2), lim = p) {
      p = j;
      iota(all(y), n - j);
      rep(i, 0, n) if(sa[i] >= j) y[p++] = sa[i] - j;
      fill(all(ws), 0);
      rep(i, 0, n) ws[x[i]]++;
      rep(i, 1, lim) ws[i] += ws[i - 1];
      for(ll i = n; i--;) sa[--ws[x[y[i]]]] = y[i];
      swap(x, y);
      p = 1;
      x[sa[0]] = 0;
      rep(i, 1, n) {
         ll a = sa[i - 1], b = sa[i];
         x[b] = (y[a] == y[b] && y[a + j] == y[b + j]) ? p - 1
: p++;
      }
   }
   rep(i, 1, n) rk[sa[i]] = i;
   for(ll i = 0, k = 0; i < n - 1; lcp[rk[i++]] = k) {
      if(k) k--;
```

```
        while(s[i + k] == s[sa[rk[i] - 1] + k]) k++;
    }
    sa.erase(begin(sa));
    lcp.erase(begin(lcp));
    return pair{sa, lcp};
}
```

## Zalgorithm.hpp

```
// Z[i] := LCP(s, s[i:])
// abacaba -> 7010301
auto Z(string s) {
    ll n = sz(s), l = -1, r = -1;
    vector<ll> z(n, n);
    rep(i, 1, n) {
        ll& x = z[i] = i < r ? min(r - i, z[i - l]) : 0;
        while(i + x < n && s[i + x] == s[x]) x++;
        if(i + x > r) l = i, r = i + x;
    }
    return z;
}
```

# algorithm

# geometry

# memo

## Primes.md

**素数の個数**

| $n$ | $10^2$ | $10^3$ | $10^4$ | $10^5$ | $10^6$ | $10^7$ | $10^8$ | $10^9$ | $10^{10}$ |
|---|---|---|---|---|---|---|---|---|---|
| $\pi(n)$ | 25 | 168 | 1229 | 9592 | 78498 | 664579 | 5.76e+6 | 5.08e+7 | 4.55e+8 |

**高度合成数**

| $\leq n$ | $10^3$ | $10^4$ | $10^5$ | $10^6$ | $10^7$ | $10^8$ | $10^9$ |
|---|---|---|---|---|---|---|---|
| $x$ | 840 | 7560 | 83160 | 720720 | 8648640 | 73513440 | 735134400 |
| $d^0(x)$ | 32 | 64 | 128 | 240 | 448 | 768 | 1344 |

| $\leq n$ | $10^{10}$ | $10^{11}$ | $10^{12}$ | $10^{13}$ | $10^{14}$ | $10^{15}$ | $10^{16}$ | $10^{17}$ | $10^{18}$ |
|---|---|---|---|---|---|---|---|---|---|
| $d^0(x)$ | 2304 | 4032 | 6720 | 10752 | 17280 | 26880 | 41472 | 64512 | 103680 |

**素数階乗**

| $n$ | 2 | 3 | 5 | 7 | 11 | 13 | 17 | 19 | 23 | 29 |
|---|---|---|---|---|---|---|---|---|---|---|
| $n\#$ | 2 | 6 | 30 | 210 | 2310 | 30030 | 510510 | 9.70e+6 | 2.23e+8 | 6.47e+9 |

**階乗**

| $4!$ | $5!$ | $6!$ | $7!$ | $8!$ | $9!$ | $10!$ | $11!$ | $12!$ | $13!$ |
|---|---|---|---|---|---|---|---|---|---|
| 24 | 120 | 720 | 5040 | 40320 | 362880 | 3.63e+6 | 3.99e+7 | 4.79e+8 | 6.23e+9 |